

IN2901

Software Development Project

TenderEase.lk

Sri Lankan Government Tendering and Bidding Platform



Group No: 30

Group Name: Pulse24

Index No

Name

234132T

N. N. Moragoda

234129P

K. G. U. Mihiranga

234198A

S. T. N. Sirisooriya

234101A

D. S. Kankanamge

234082M

H. G. C. S. Hulangamuwa

Bachelor of Science Honors in Information Technology (BScHons (IT))
Faculty of Information Technology
University of Moratuwa

Table of Contents

Executive Summary	3
List of Acronyms and Abbreviations	4
1. System Overview	5
2. Objectives and Aims	6
2.1 Objective	6
2.2 Aims	6
3. Core Users & Roles	7
4. High-Level Features (MVP)	8
5. Functional Requirements	9
5.1 Vendor Module	9
5.2 Vendor / Supplier Organization Details	10
5.3 Details of the Authorized Officer of the Vendor/Supplier	11
5.4 Details of the Liaison Officer	12
6. System Flow Overview	13
6.1 Procurement Publishing	14
6.2 Vendor Interaction and Quotation Submission	14
6.3 Transparent and Guided Process	15
6.4 ER Diagram	15
7. Technologies Used	16
7.1 Frontend Tech Stack	16
7.2 Backend Tech Stack	17
7.3 Storage	17
8. Novelty and Innovation	18
9. Work Allocation	19
9.1 Tender Publishing & Master Data Setup	19
9.2 Bidder Portal & Vendor Management	20
9.3 Clarifications & Addenda	20
9.4 Bid Opening & Evaluation	21
9.5 Award, Compliance & Reports	21
9.6 Shared Responsibilities	22
10. Expected Outcomes and Benefits	22

11. Conclusion	23
References.....	24

Executive Summary

TenderEase.lk is a centralized e-procurement platform built to digitize and streamline the Sri Lankan government's tendering and bidding process. It connects **government departments and suppliers** in one secure and transparent online space, making the entire process- from publishing tenders to awarding contracts faster and easier.

While other platforms already exist mainly lists tender notices for sharing, TenderEase.lk manages the entire tendering process. Departments may create, publish, and manage tenders, while vendors can register in the system, submit tenders electronically, communicate with departments on requirements for bids, and review everything via the system. Activities are recorded automatically, ensuring transparency and lessening possible manual interference.

In the first phase, TenderEase.lk intends to provide services to six major government departments: **Agriculture, Education, Technology, Health, Finance and Transportation**. It will strive to save time and diminish errors and increase transparency of procuring by automating several procedures and enforcing clear approval workflows.

Overall, the system helps bring Sri Lanka's public procurement process closer to international digital standards **improving efficiency, accountability, and trust between the government and its suppliers**.

List of Acronyms and Abbreviations

Acronym/Term	Description
CPV	Common Procurement Vocabulary (used for classifying procurements)
ER Diagram	Entity-Relationship Diagram (a structure model used to represent the database design)
ICB	International Competitive Bidding (a government procurement method)
IFeQ	Invitation for e-Quotation (published by the procuring entity)
MVP	Minimum Viable Product (the initial version of the system with core features)
NCB	National Competitive Bidding (a government procurement method)
NIC	National Identity Card (used for official identity verification)
RFQ	Request for Quotations (a government procurement method)

1. System Overview

TenderEase.lk is developed as an end-to-end digital solution for administering government tendering and bidding within Sri Lanka. It joins two important user groups **Procuring Entities** like ministries and departments and **Vendors or Suppliers** within one secure and user-friendly platform. The system supports all stages of the procurement cycle, right from publishing tenders and clarifications to bid receiving, evaluation, and contract awarding.

At its core, TenderEase.lk aims to **reduce paperwork, eliminate delays and create a transparent and traceable workflow**. Each action in the system is logged with a timestamp, ensuring accountability and auditability at every step. The system automatically reviews vendor registrations based on uploaded documents, making the approval process faster and more reliable.

The other hand offers vendors a quick registration process, viewing of active and engaged procurements, uploading of quotation, certificate, and specification documents in digital format, amongst other services; while on the Procuring Entities' side, officers can create tenders, handle vendor clarifications, and manage evaluations and approvals all from one secure dashboard.

TenderEase.lk also enforces government procurement standards by supporting methods such as National Competitive Bidding (NCB), International Competitive Bidding (ICB), and Request for Quotations (RFQ). In addition, it ensures every tender follows a structured flow from notice publication and clarification to bid opening, evaluation, and final award.

By **centralizing** all tender-related activities, TenderEase.lk improves coordination between departments and vendors while minimizing manual intervention. The result is a faster, fairer, and more consistent procurement process that aligns with Sri Lanka's vision for a transparent and technology-driven public sector.

2. Objectives and Aims

2.1 Objective

The main objective of TenderEase.lk is to create a transparent, efficient, and fully digital platform for managing the government tender and bidding process in Sri Lanka. The system aims to simplify procurement for both government entities and suppliers while ensuring that every step follows the official procurement guidelines and maintains public trust.

2.2 Aims

- **Enhance Transparency:** Make all tender-related information easily accessible and traceable, ensuring every action is recorded for audit and public review.
- **Save Time and Reduce Errors:** Cut down the tender cycle time by automating repetitive tasks, minimizing paperwork, and avoiding human mistakes.
- **Ensure Fair Competition:** Provide equal access to tender information for all registered vendors through automated notifications and standardized documentation.
- **Strengthen Compliance:** Enforce government procurement rules, timelines, and approval hierarchies within the system.
- **Improve Accountability:** Record every action from tender creation to bid evaluation with digital timestamps and audit trails.

3. Core Users & Roles

The TenderEase.lk platform is designed to serve multiple user groups within the government procurement process. Each role has specific responsibilities to ensure smooth, transparent, and compliant tender management.

- **Procurement Admin (Ministry/Department):** Creates and publishes tenders, manages clarifications, and coordinates the overall procurement process.
- **Evaluation Committee Member:** Reviews and scores bids based on technical and financial criteria, contributing to the final evaluation report.
- **Approving Authority (Head of Department/Secretary):** Reviews recommendations and provides final approval for tender awards.
- **Bidder (Vendor/Supplier):** Registers on the platform, submits bids, uploads required documents, and communicates with the procuring entity.
- **Auditor/Oversight Officer:** Monitors system logs, reviews procurement records, and ensures compliance with government procurement guidelines.
- **System Administrator:** Manages platform operations, user access, and ensures data security and system reliability.

4. High-Level Features (MVP)

The initial version of **TenderEase.lk** focuses on delivering key features that support the entire tender and bidding process while maintaining transparency, efficiency, and compliance.

1. Tender Publishing

- Departments can create and publish tenders with standard fields and upload official documents.
- Automated tender notices, schedules, and checklists are generated for consistency.

2. Vendor Portal

- Vendors can register, upload business documents, and submit bids online.
- Secure bid submission with time-based locking before deadlines.

3. Clarifications & Addenda

- Vendors can raise questions, and responses are shared with all registered bidders.
- Any changes or corrections are versioned and auto notified to users.

4. Bid Opening & Evaluation

- Automatic bid opening at the scheduled time with digital attendance logs.
- Evaluation based on technical and financial criteria with scoring and review tools.

5. Award & Notification

- System-generated recommendation and approval chain.
- Automatic notifications to winning and unsuccessful bidders.

6. Audit & Compliance

- Complete audit trail of every action with timestamps and digital signatures.
- Secure storage of all procurement records for review and accountability.

5. Functional Requirements

5.1 Vendor Module

Vendor Registration

The Vendor Registration section enables suppliers or service providers to register in the government tendering system. This module collects essential business and financial information to verify the authenticity and eligibility of vendors before they can participate in tenders.

Documents

Field	Description
Business Registration Document	Upload a valid business registration certificate issued by the authorized body.
Bank Statement or Audit Report	Provide a recent bank statement or audited financial report to verify financial stability.
Tax Report	Submit the latest tax clearance or tax payment report to confirm compliance with tax regulations.
Proof of Past Experience (Optional)	Attach documents or certificates showing previous contracts or project experience.

5.2 Vendor / Supplier Organization Details

Field	Description
Business Registration Authority	Name of the government body that issued the business registration.
Business Name (Uppercase)	Official name of the company or organization, entered in uppercase letters for consistency.
Country	The country where the business is legally registered and operates.
Business Registration No	Unique identification or registration number issued by the authority.
Type of Organization	Nature of the entity (e.g., Sole Proprietorship, Partnership, Private Limited, Public Limited).
Registered Address	Official address where the business is registered.
City	The city in which the business is located.
Province	The province or region of the business's registered address.
Website	Official website link of the vendor or supplier, if available.
Official Mail	Primary official email address for communication and notifications.
Official Telephone No	Main contact number of the business for verification or inquiries.

5.3 Details of the Authorized Officer of the Vendor/Supplier

This section records the personal and contact details of the individual officially authorized to represent the vendor or supplier in all procurement-related communications and transactions.

Field	Description
NIC / Passport No	National Identity Card number or Passport number of the authorized officer for identity verification.
Name	Full name of the authorized officer as per official identification.
Designation	Job title or position of the officer within the organization (e.g., Director, Manager, Procurement Officer).
Mobile Phone No	Personal mobile number of the authorized officer for direct communication.
Email	Official email address of the authorized officer for correspondence and system notifications.

5.4 Details of the Liaison Officer

This section captures the information of the designated Procurement Liaison Officer, who serves as the primary point of contact between the procuring entity and vendors throughout the tendering process.

Field	Description
Title	The formal title of the liaison officer (e.g., Mr., Ms., Dr., Eng.).
Procurement Liaison Officer	Full name of the officer appointed to manage procurement communications.
Designation	Official position or role of the liaison officer within the organization.
NIC	National Identity Card number of the liaison officer for identification and verification.
Mobile No	Official mobile contact number for direct communication regarding procurement matters.
Email	Official email address used for all procurement-related correspondence and notifications.

6. System Flow Overview

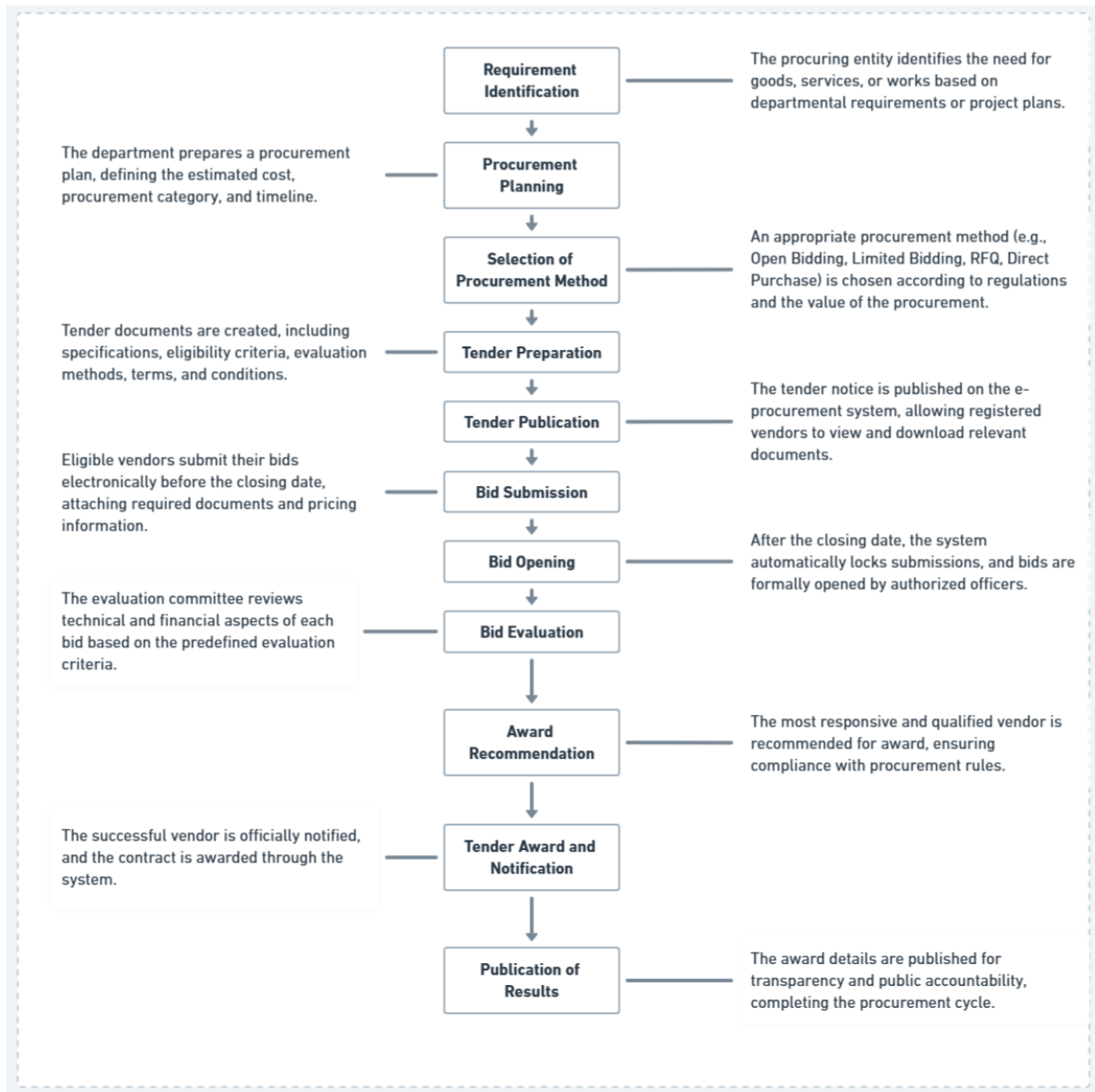


Figure 6.1 – System Flow Diagram

<https://whimsical.com/2QWJHFWnj9xL99wedgGswz>

6.1 Procurement Publishing

The process begins when a government **ministry or department** publishes a new procurement through the system.

Key steps include:

- Entering **Primary Details** such as Ministry, Department, Procurement Method, and Category.
- Selecting a **CPV (Common Procurement Vocabulary)** code for classification.
- Entering **Quotation Number, Title, Estimation, and Funding Source**.
- Publishing the **Invitation for e-Quotation (IFeQ)** with start, closing, and opening dates.
- Selecting a **Process Template** (standard or custom), which provides **guidance and instructions** based on the selected method.
- Adding **Further Details**, including the **appointment of evaluation committees** and other administrative data.

6.2 Vendor Interaction and Quotation Submission

Once a tender is published, **registered vendors** can view and participate in the process through their dashboards.

Vendors can:

- View **Engaged and Active Procurements** they are eligible for.
- **Review IFeQ details** and send **clarifications or queries** to the procuring entity.
- **Prepare and submit their quotations** digitally.
 - Upload **Commercial Requirements** documents.
 - Attach **Certificates** (such as business registration or compliance).
 - Upload **Indications of Specifications and Bid Documents**.
 - Complete and **finalize their quotation submission** before the deadline.

6.3 Transparent and Guided Process

Throughout the flow, the system ensures:

- Clear **guidance** at each step to reduce procedural errors.
- Secure and time-stamped submissions to maintain **integrity and fairness**.
- Simplified **review and communication channels** between vendors and procuring entities.

6.4 ER Diagram

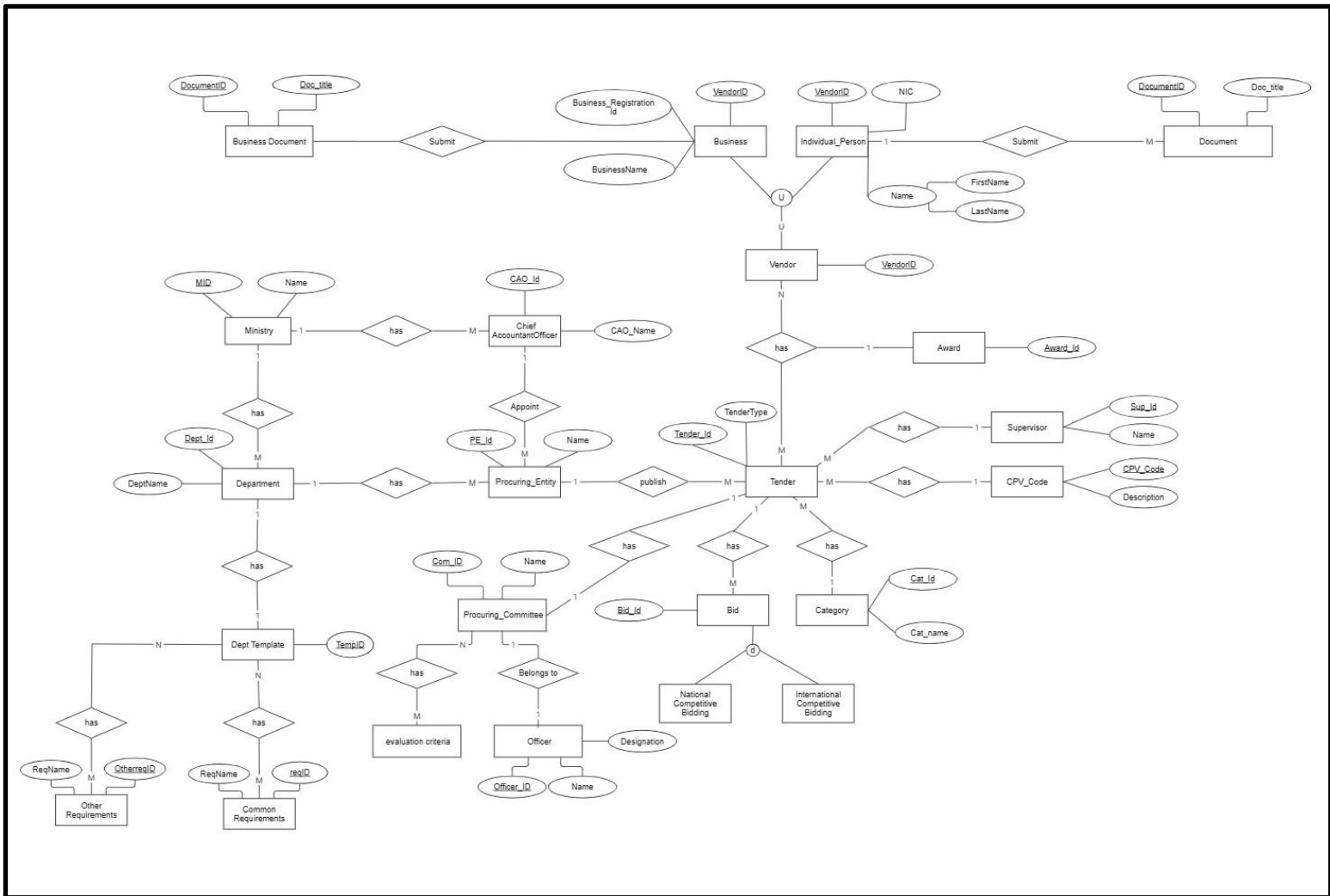


Figure 6.2 – ER Diagram

https://miro.com/app/board/uXjVJ5uNCTo=/?share_link_id=971636692078

7. Technologies Used

7.1 Frontend Tech Stack

Area	Our Choice	Pros	Cons
Frontend Framework	Next.js	Huge ecosystem, SEO, Hybrid rendering	Slightly heavier than Remix/SvelteKit, learning curve for App Router
UI + Styling	Tailwind + shadcn/ui	Tailwind → utility-first (pre-defined classes), scalable Shadcn → accessible, clean components, highly customizable with tailwind	Slightly more setup work, opinionated
Data Fetching & State	Zustand (for global state)	Zustand → super minimal, fast Easy to use than other libraries and community is bigger than other libraries except Redux toolkit. But it is easier to use than Redux toolkit.	Zustand lacks opinionated structure vs Redux
Form Validation	Zod	Zod → TS-first validation, New but old enough to stable. Huge community than Yup.	Zod is still evolving compared to Yup. Yup was originally created for js.
HTTP Client	Axios	Interceptors, widely used	Slightly heavier than Fetch
Form Handling	Formik	Lightweight + TS-first validation	More boilerplate than Formik in simple forms
Internationalization	i18n, next-intl	Built for Next.js, SSR-friendly, minimal boilerplate	Still maturing, smaller ecosystem than i18next

Unit Tests	Vitest + RTL	Mature ecosystem, widely adopted, works seamlessly with Next.js	Slightly slower than Vitest in larger projects
-------------------	-----------------	---	--

7.2 Backend Tech Stack

Area	Our Choice	Pros	Cons
Backend framework	Spring Boot V3. 5. 6 (Java V21)	Mature, enterprise-grade, huge ecosystem, strong type safety	Verbose compared to Node/Go: slower cold starts
Authentication	JWT, Auth 2.0		
Caching & Queues	Redis and RabbitMQ		
Database	PostgreSQL		

7.3 Storage

Database – Postgre SQL

Object Storage – AWS S3

8. Novelty and Innovation

1. **Automated Vendor Registration Review**

The system automatically validates vendor registration details and uploaded documents using AI-based verification techniques. This reduces the need for manual review and ensures only verified vendors proceed to the procurement stages.

2. **AI-powered chatbot that supports Conversational Search**

Apart from typical search by category, value, location, and closing date, we are implementing a chatbot that supports Conversational Search. A natural language chatbot that allows users to simply ask, "Are there any new IT support tenders in the Western Province?"

3. **Context Bridge (Drag and Drop)**

A user can perform a traditional search and see a list of tenders and They can then **drag one or more tenders** from the search results directly into the chatbot window. Once a tender is "dropped" into the chat, the user can immediately use pre-built AI tools or ask specific questions.

4. **AI Ranking and Recommendation System**

Vendors are ranked based on historical performance, reliability, pricing trends, and compliance levels. The system uses AI to recommend the most suitable vendors for each procurement, improving the quality and fairness of the selection process.

5. **Smart Procurement Matching**

Leveraging machine learning, the system intelligently matches new tenders with the most relevant vendors based on their business categories, past participation, and specialization. This ensures efficient tender distribution and higher vendor engagement.

6. Automated Compliance Scoring

Before final submission, each quotation or bid is automatically checked for completeness and compliance against the tender's requirements. The system generates a compliance score, enabling procurement officers to quickly identify valid and high-quality submissions.

7. AI Document Summarizer

A built-in AI assistant summarizes lengthy tender documents and vendor proposals, providing concise overviews for procurement officers. This significantly reduces the time required for document analysis and decision-making.

8. Auto generation Tender Notices

9. Work Allocation

9.1 Tender Publishing & Master Data Setup

(234132T – N. N. Moragoda)

Core Functional Work	Extra Responsibilities
Build tender creation flow: Goods/Works/Services, NCB/ICB, dates, etc.	Own master data setup (categories, agencies, codes like CPV/NIGP).
Upload and store SBDs, agency templates.	Design DB schema for tenders, documents, meetings, categories.
Auto-generate tender notice/checklist.	Create validation rules (mandatory fields, date ranges, etc.).
Schedule clarifications/pre-bid meeting with calendar integration.	Implement file versioning for tender documents.
	Automated Vendor Registration Review

9.2 Bidder Portal & Vendor Management

(234082M - H. G. C. S. Hulangamuwa)

Core Functional Work	Extra Responsibilities
• Vendor registration & KYC form (with category tagging).	• Build vendor profile management (update KYC, blacklist/whitelist vendors).
• Bid submission (upload docs, forms, BoQ, pricing).	• DB schema for vendors, kyc_docs, submissions, submission_files.
• Submission lock after deadline; generate proof of submission.	• Implement unique vendor ID & duplicate prevention logic.
	• Handle email/SMS confirmations for submissions.
	• AI-powered chatbot that supports Conversational Search

9.3 Clarifications & Addenda

(234129P - K. G. U. Mihiranga)

Core Functional Work	Extra Responsibilities
• Question/answer window for bidders - agency replies.	• Build notification service (email/SMS trigger when clarifications/addenda are posted).
• Publish clarifications visible to all bidders.	• DB schema for clarifications, clarification_replies, addenda_versions.
• Addenda/corrigenda publishing with version history & auto notifications.	• Implement document version control (hash + timestamp check).
	• Manage audit logs for all Q&A and addenda changes.

9.4 Bid Opening & Evaluation

(234101A - D. S. Kankanamge)

Core Functional Work	Extra Responsibilities
Time-boxed bid opening; attendance log.	Build evaluation criteria setup module (pass/fail, weights, minimum scores).
Multi-criteria evaluation (technical + financial).	DB schema for opening_sessions, attendance, evaluations, scores.
Weighted scoring models with consensus sheet generation.	Implement committee workspace with role-based access (reviewer vs chair).
	Generate evaluation reports (PDF/Excel).

9.5 Award, Compliance & Reports

(234198A - S. T. N. Sirisooriya)

Core Functional Work	Extra Responsibilities
Recommendation note & approvals chain.	Implement post-award disclosure module (public access: winning bidder, contract value, justification).
Award letter generation & notifications (winning + unsuccessful bidders).	DB schema for awards, approvals, notifications, reports.
Reports & dashboards: KPIs (cycle time, SME participation, award values).	Build dashboard APIs (cycle time, bidder count, participation trends).
	Manage compliance rules (statutory retention, post-award data visibility).

9.6 Shared Responsibilities

- **Auditability:** Immutable logs (who/what/when), timestamps.
- **Security:** RBAC, MFA, field-level encryption.
- **Search & Alerts:** Keyword/category search, email/SMS alerts.
- **Statutory Compliance:** Retention rules applied to their domain's DB tables.

10. Expected Outcomes and Benefits

The proposed **TenderEase.lk** system is expected to transform the traditional tendering process into a fully digital, transparent, and efficient ecosystem. By integrating automation and AI-driven insights, the platform delivers the following key outcomes and benefits:

1. **Increased Transparency:** Every stage of the procurement process is digitally recorded, ensuring full accountability and auditability.
2. **Faster Processing:** Automated document validation and evaluation significantly reduce manual delays.
3. **Fair Vendor Selection:** AI ranking and compliance scoring promote fairness and objectivity in tender awards.
4. **Reduced Administrative Workload:** The system minimizes paperwork, data duplication, and human intervention.
5. **Enhanced Decision-Making:** AI recommendations and summarized reports support data-driven procurement decisions.

6. **Better Vendor Engagement:** Smart matching and clear communication channels encourage more qualified vendors to participate.
7. **Cost Savings:** Efficient workflows and reduced manual errors lower overall operational costs for government departments.
8. **Data-Driven Insights:** Centralized data allows departments to track performance trends and optimize procurement strategies.
9. **Secure and Scalable System:** Cloud-based infrastructure ensures data integrity, accessibility, and scalability across ministries.
10. **Public Trust and Credibility:** Transparent operations strengthen public confidence in the government's procurement process.

11. Conclusion

The **TenderEase.lk** platform aims to modernize the government tendering and bidding process through smart automation, transparency, and AI innovation. By bridging the gap between vendors and government departments, it ensures a fair, efficient, and future-ready procurement system that sets a new digital standard for public sector operations.

References

Theekshana, “Promise.” <https://promise.lk/>

“eGP System Sri Lanka,” *YouTube*. <https://www.youtube.com/@PromiseSriLanka>

“Ministry of Finance - Sri lanka.” <https://www.treasury.gov.lk/web/procurement-guidelines-and-manuals/section/procurement%20manual>

Kalinga Gananath, “Procurement Process in Sinhala / ප්‍රසම්පාදන ක්‍රියාවලිය - සිංහලෙන්,” *YouTube*. Oct. 08, 2022. [Online]. Available: <https://www.youtube.com/watch?v=YrMVWT6cMcA>

“eRoc.” <https://eroc.drc.gov.lk/home/company/verify>